IT Applications in Business

Blockchain and cryptocurrencies

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HR EXCELLENCE IN RESEARCH

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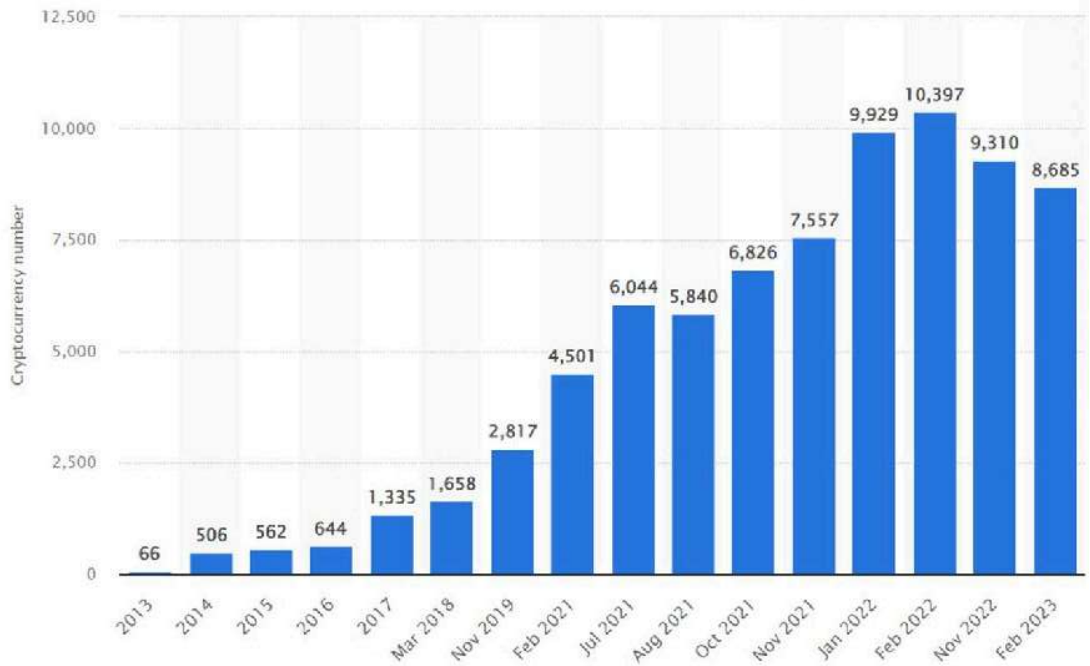
Lecture 2: ***Blockchain and Cryptocurrencies***

1. **Cryptocurrencies**
   - Bitcoin, Ethereum, Ripple
   - History of payments systems
   - BTC vs. fiat money
2. **Process of „mining"**
3. **Law and regulations**
4. **Cryptocurrencies Wallets**
5. **BTC scaling issue**
6. **ICO & DAO**
7. **Consensus mechanisms**
8. **Smart contracts**
9. **Blockchain applications**

Cryptocurrencies

- digital assets
- use medium of exchange (commodities)
- facilitate transactions
- use cryptography to ensure
  - **security** and **validity** of transactions



- not backed by a central authority (e.g., government) that could determine its value
- value determined proportionally to interest in cryptocurrency and work behind it (mining)

Cryptocurrencies – three main concepts

- Public ledgers
  - holds info about ALL transactions
  - non controllable by a single individual or agency
- Mining
  - validates the transactions
  - updates ledgers
- Transactions

Cryptocurrencies

- cryptocurrency is **usually** *created* / “issued” trough a processed called *mining*
- a digital form of *mining* (in former times eg. gold nuggets)
- some (e.g. BTC) have finite amount
- usually free / opensource to use

Brief history of payment system

- Commodity money
- Representative money
- FIAT money
- Checks
- Electronic payment – wire transfers
- Electronic payment – debit cards

Brief history of payment system

- 9000-6000 BC - shells, livestock and plant products
- 3000 BC - Mesopotamia, Shekel=180 Grains of Barley (11g)→unit of weight and currency
- 1000 BC - Zhou dynasty, first standardized coinage (knives / spades)
- 700-500 BC - Aegean coins
- 7th century (AC) – China first banknotes
- 1661 – Bank of Sweden first European banknote
- 1971 – Richard Nixon removed backing of gold from USD

Bitcoin

- Digital currency – not backed by a single authority
- Decentralized and distrusted
- Not issued, but mined ➔ process of verifying transactions validity
  - Similar method to physical mining of gold
- New coins are generated as rewards for verifying a certain number of transactions
- Creation rate is automatically halved evert few years as more BTC are generated
- Max. number of BTC limited to 21 milion (will be mined by est. 2140)

Bitcoin

- Smallest denomination of BTC is 1 Satoshi = 0,000 000 01 BTC
- Public ledger
- Exchange
  - Facilitates sending and receiving
- Wallets
  - Stores BTC

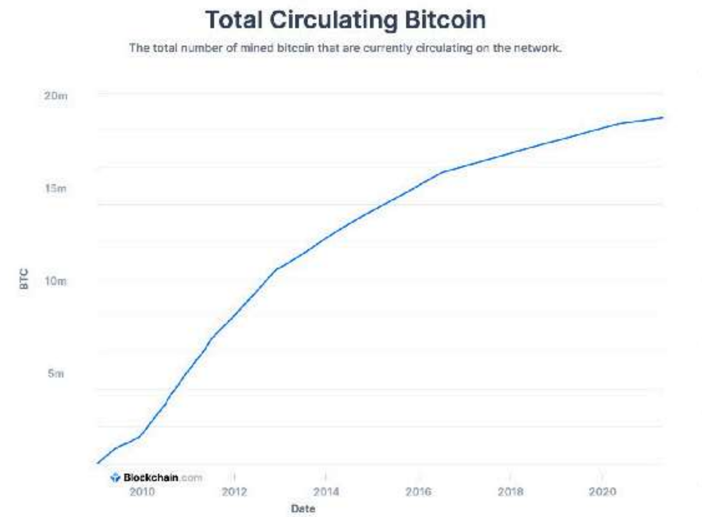
Bitcoin

- 144 blocks a day are mined on average, and there are 3,125 BTCs / block

144 x 3,125 = 450 BTC per day

- because many miners are adding new ***hashpower***, over the last few years blocks have often been found at 9,5-minute intervals rather than 10 this creates new bitcoins faster, so on most days there are more than 900 new bitcoins created

Total BTC in Existence	19 856 778,125
Bitcoins Left to Be Mined	1 143 221,9
% of Bitcoins Issued	94,556 %
New Bitcoins per Day	**450**
Mined Bitcoin Blocks	894 169



https://www.buybitcoinworldwide.com/how-many-bitcoins-are-there/

Bitcoin vs. fiat currency

- ***FIAT money*** a currency without intrinsic value
  - established as money, often by government regulation
  - issued by a government
  - value determined by a central authority
  - not very divisible
  - easy to transact, can be used in a real life
  - not scarce – can be issued (almost) indefinitely
  - widely accepted as a medium for exchange

Bitcoin vs. fiat currency

- ***FIAT money***
  - not finite
  - introduced as an alternative to
    - commodity money (gold, silver)
    - representative money (backed by gold)

Bitcoin vs. fiat currency

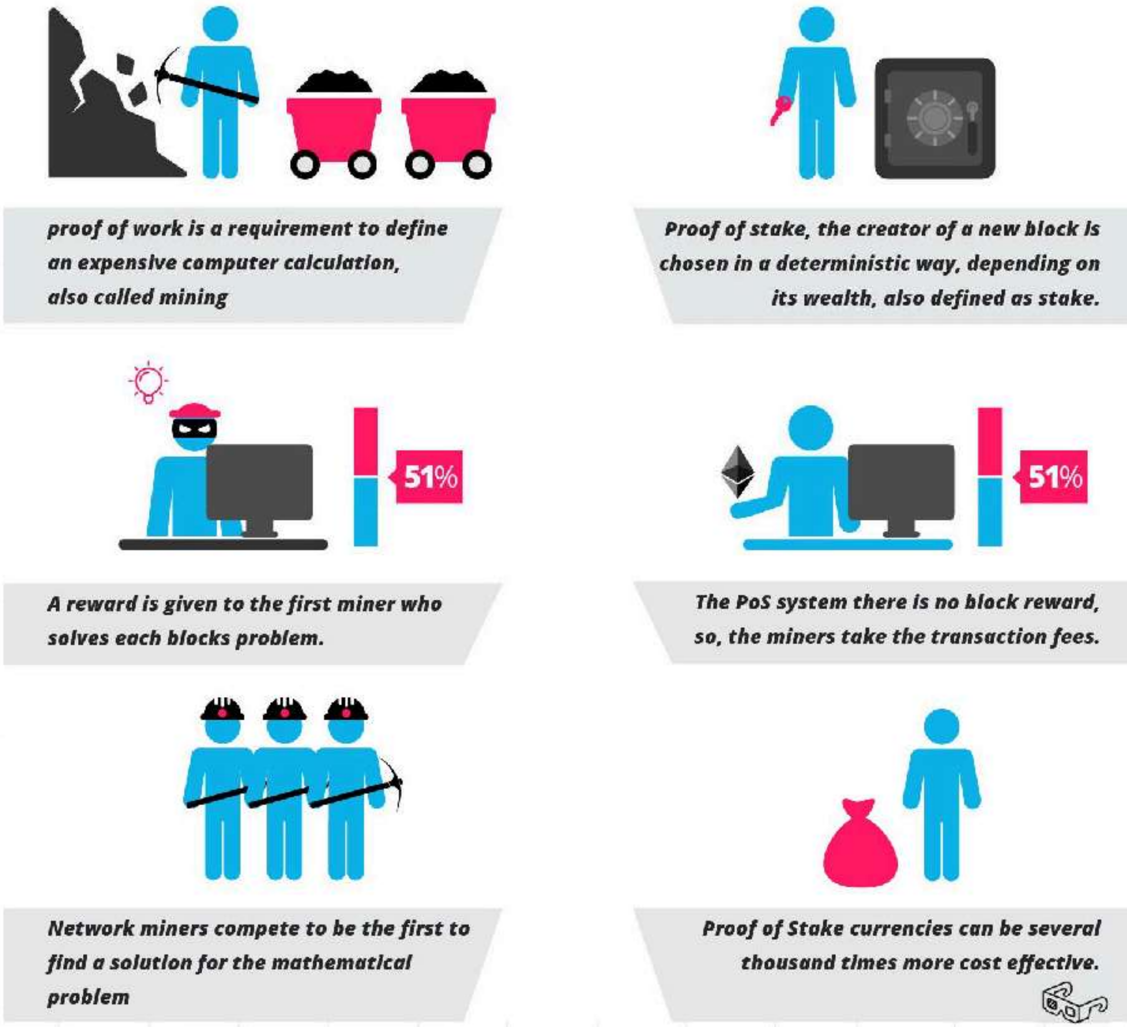
- ***Bitcoin –*** *a cryptocurrency and worldwide payment system, essentially P2P network*
  - decentralized and distrusted (controlled by a network of users)
  - highly divisible
  - very secure (validity needs to confirmed by a number of users)
  - easy to transact (not every vendor accepts BTC)
  - wide acceptation (?)
  - value determined by the community and NOT by a single authority

Bitcoin vs. fiat currency

- ***Bitcoin –*** *a cryptocurrency and worldwide payment system*
  - free to use
  - requires internet connection
  - not in real life presence

PoW vs. PoS

Proof of Stake	Proof of Work
Block creators are called ***validators***	Block creators are called **miners**
Participants must own coins or tokens to become a validator	Participants must buy equipment and energy to become a miner
Energy efficient	Energy inefficient
Security through community control	Robust security due to expensive upfront requirement
Validators receive transactions fees as rewards	Miners receive block rewards



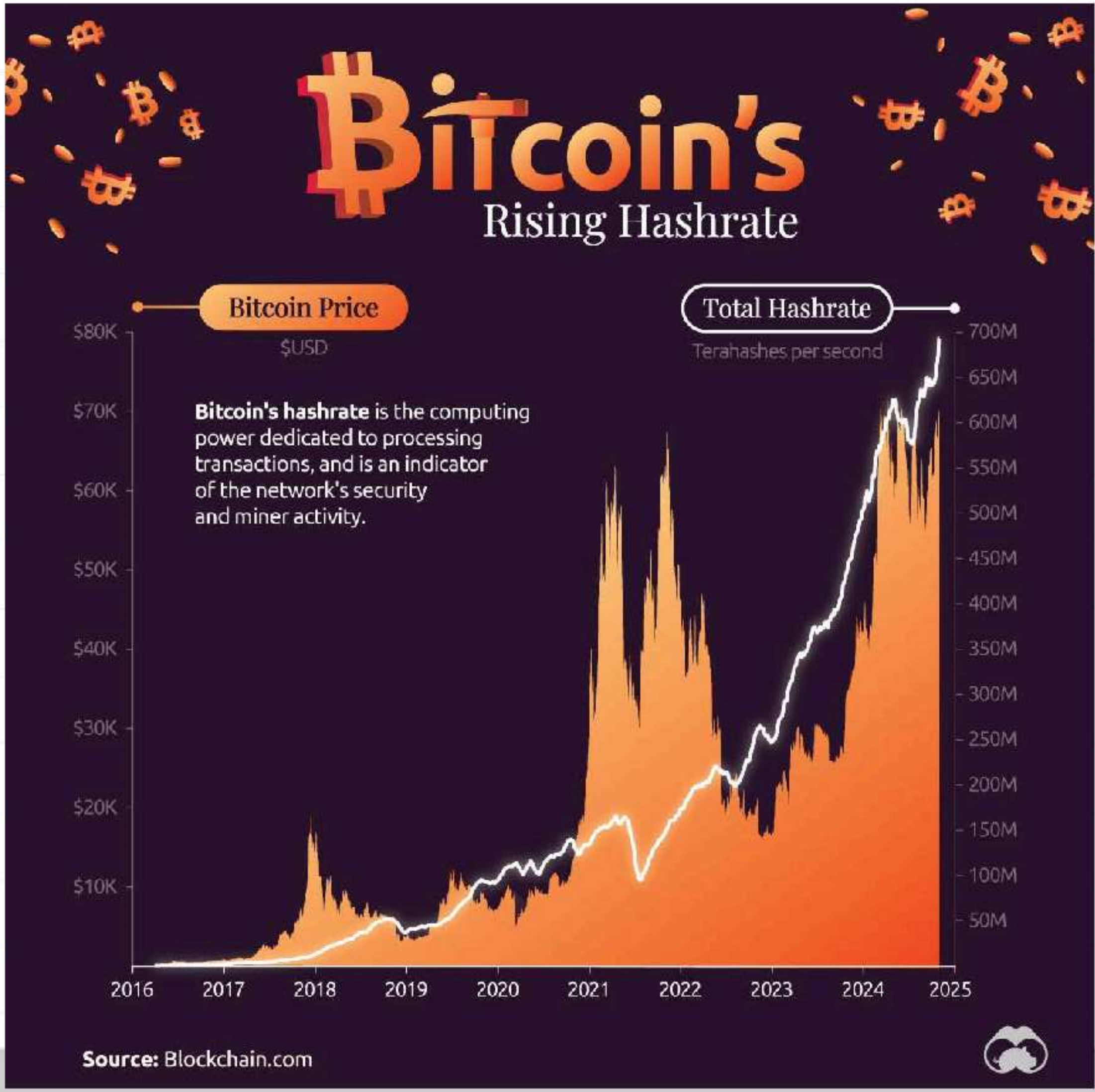
Source: Blockgeeks

Process of mining

- ***Bitcoin –*** *a cryptocurrency and worldwide payment system*
  - special software to solve math problems (verification of transactions)
  - miners are rewarded with fracture of BTCs for solving the math problems
  - verification and approval of BTC transactions
  - incentives attract more Miners
  - more miners make the system more robust and secure
  - wider adoption by the public

Process of mining

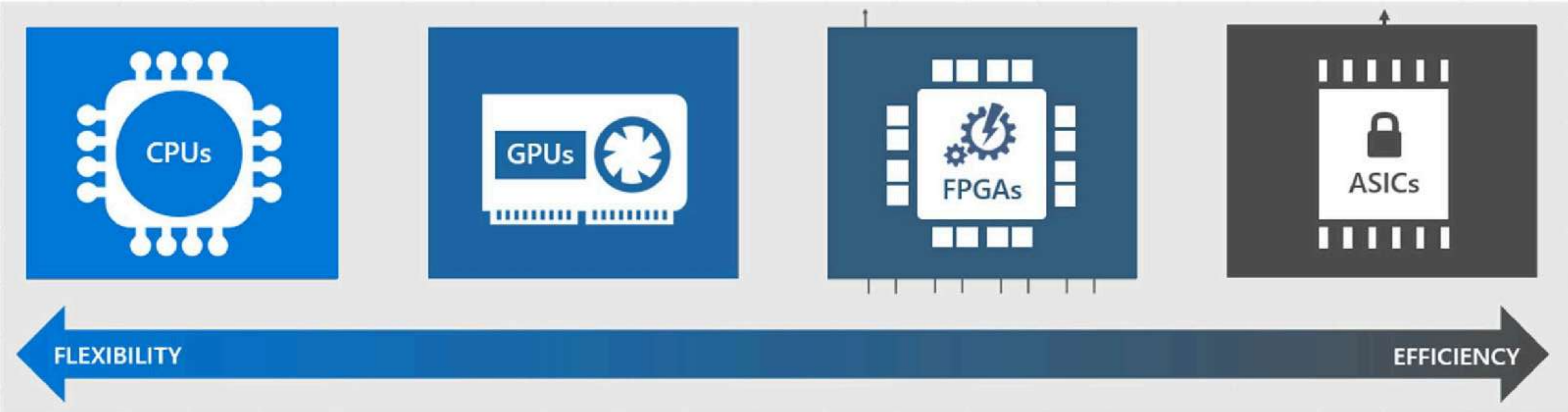
- ***Bitcoin***
  - BTC network automatically changes the difficulty of math problems to prevent too quick depletion of total BTCs that can be issued
  - in the past ***CPU*** were sufficient to cope with math problems – now not a single ***GPU*** is good enough
  - ***ASIC*** (*Application Specific Integrated Circuit* chips) – far more suitable for mining
    - dedicated for specific tasks, energy efficient


Bitcoin's
Rising Hashrate
Bitcoin Price
$USD
Total Hashrate
Terahashes per second
Bitcoin's hashrate is the computing power dedicated to processing transactions, and is an indicator of the network's security and miner activity.
$80K
$70K
$60K
$50K
$40K
$30K
$20K
$10K
700M
650M
600M
550M
500M
450M
400M
350M
300M
250M
200M
150M
100M
50M
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Source: Blockchain.com

Process of mining

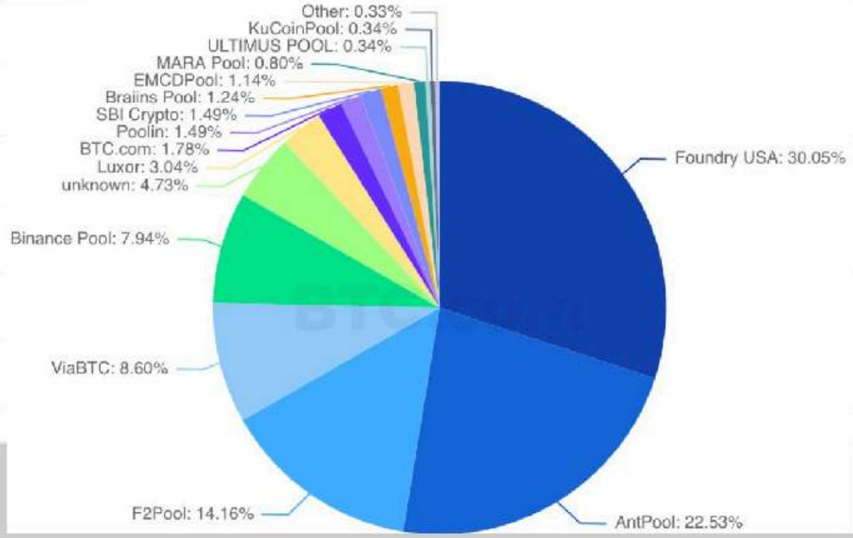
- ***Bitcoin***
  - BTC network automatically changes the difficulty of math problems to prevent too quick depletion of total BTCs that can be issued
  - **CPU**s – inefficient (too slow)
  - **GPU**s – expensive, energy inefficient
  - **ASIC**s – unreasonably expensive, dedicated for mining

Process of mining



Process of mining

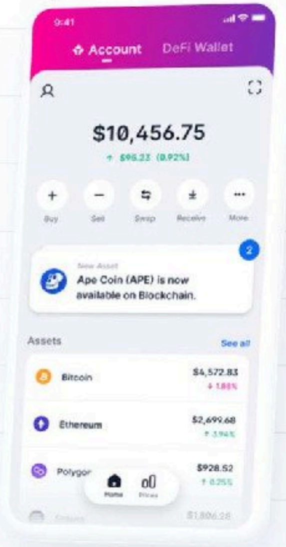
- ***Mining Pool(s)***
  - collection of miners to mine coins faster
  - coins are rewarded and split according to individual contribution



What is a wallet?

- ***Wallet***
  - equivalent of a bank account
  - a collection of private keys that facilitate of
  - sending / receive / store cryptocurrencies
  - allow to store different type of cryptocurrencies
- ***Type of wallets***
  - software online
  - physical hardware (store private key on a physical device e.g. USB stick)

Wallet – cold vs. hot



- ***Hot*** – a wallet that is connected to the internet
  - free
  - easier to setup, access
  - accept more tokens
  - more susceptible to hackers, possible regulation, and other technical vulnerabilities

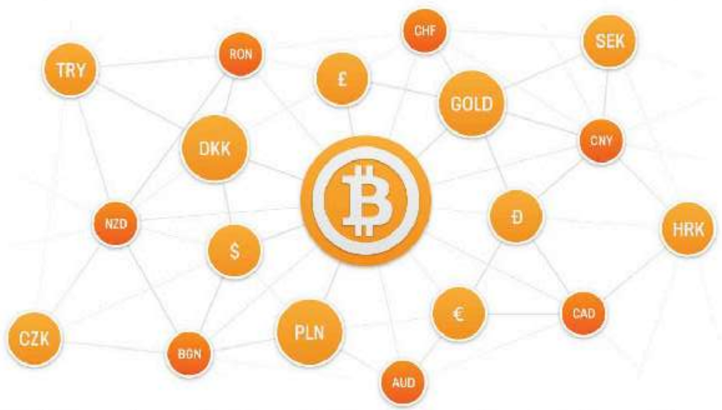
Wallet – cold vs. hot

- ***Cold*** *– a* wallet that IS NOT connected to the internet
  - more secure
  - Expensive (80+ EUR)
  - don't accept many cryptocurrencies
  - Air-gapped computers can be compromised*

- shop.trezor.io → BTC, BCH, BTG, ETH, ZCash, Dash
- www.ledgerwallet.com → BTC, BCH, BTG, ETH, ZCash, Ripple, Dash, ARK, Stellar

What is an exchange?

- ***Exchange*** *allows conversion of one currency to another*
  - many exchanges act as wallets as well
  - act similarly to banks (store and restore currency only when is it open / available)
  - allows exchange various currencies



Value of cryptocurrencies

- Neither central authority, nor organization shapes Crypto value
- Supply & demand market rules
  - More people want it → higher market value
  - Supply steadily increases and demand as well → value rises
- Values of Cryptocurrencies is reputation based
  - Better perception → higher value

How to use cryptocurrencies

- An exchange account / wallet
- Public address (e.g., email)
- Buy it (*fiat* currency or another cryptocurrency)
- Mine for BTC
- Apply for a debit card
- Spend it online / offline (physical store)
- Trade BTC (like stocks / bonds) – make money from the difference

Illicit activities

Największa kopalnia kryptowalut w Polsce zamknięta. Zarzuty wyłudzeń na co najmniej 4,5 mln zł

- Dark web
  - Acquiring illegal / stolen information data
- Dealing with drugs / weapons
- Ransom
  - Kidnapping
  - Cyber attacks

"Pay $20,000 in Bitcoin, or a Bomb will detonate in your building"

Good day. My recruited person has hidden an explosive device (Tetryl) in the building where your company is located. It is assembled according to my instructions. It can be hidden anywhere because of its small size, it is impossible to destroy the building structure by my explosive device, but in case of its explosion there will be many wounded people.
My mercenary keeps the territory under the control. If he sees any unnatural behavior, panic or policemen the bomb will be exploded.
I can call off my recruited person if you pay. You pay me 20'000 usd in Bitcoin and the device will not explode, but do not try to fool me -I warrant you that I will call off my recruited person only after 3 confirmations in blockchain network.

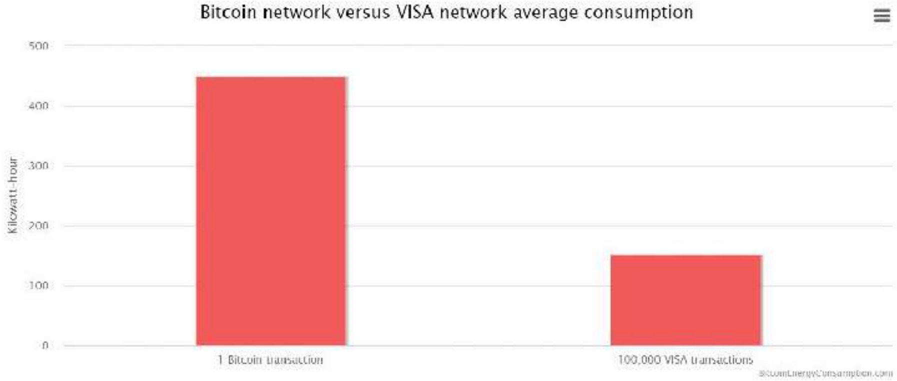
My payment details (BTC address): 15qH84uLC49CmC6jRE958Qjcf9WRZ2rMuM

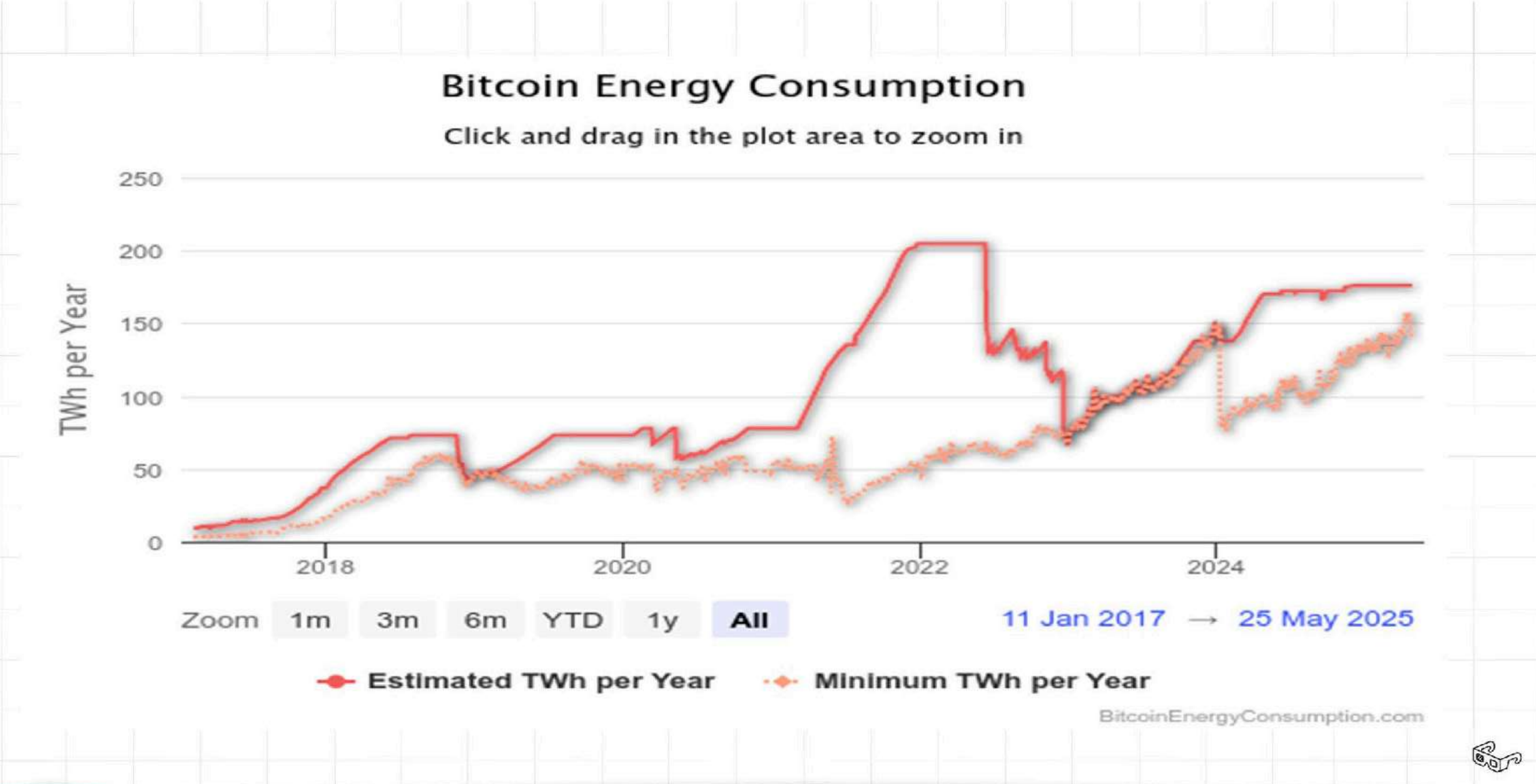
You must solve problems with the payment by the end of the workday, if the workday is over and people start leaving the building the bomb will explode.
Nothing personal this is just a business, if you don't send me the bitcoin and an explosive device detonates, other companies will send me more money, because this isnt a one-time action.
For security and anonymity reasons, I wont enter this email. I monitor my Bitcoin address every 35 minutes and after receiving the transaction I will give the command to my person to leave your area.

If an explosion occurred and the authorities read this letter!
We arent terrorists and dont take any responsibility for explosions in other buildings.

Ups & Downs

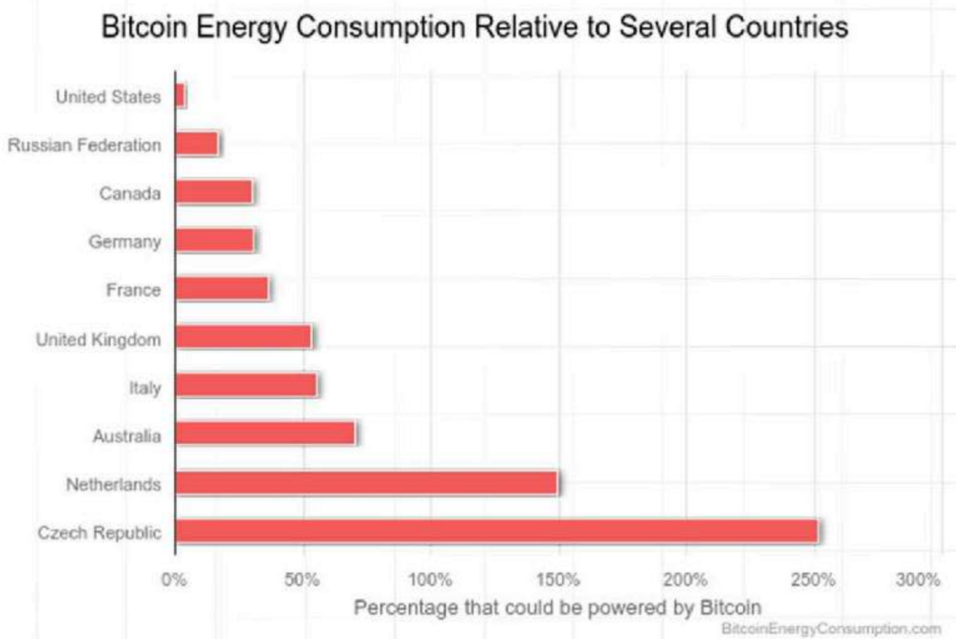
- Nascent technology
- Uncertain regulatory status
- Large energy consumption
- Control, security and privacy
- Integration concerns
- Cultural adoption
- Cost
- Challenges associated with audit, taxes, and compliance




Bitcoin Energy Consumption
Click and drag in the plot area to zoom in
250
200
150
100
50
0
TWh per Year
2018
2020
2022
2024
Zoom 1m 3m 6m YTD 1y All
11 Jan 2017 → 25 May 2025
Estimated TWh per Year
Minimum TWh per Year
BitcoinEnergyConsumption.com

Ups & Downs

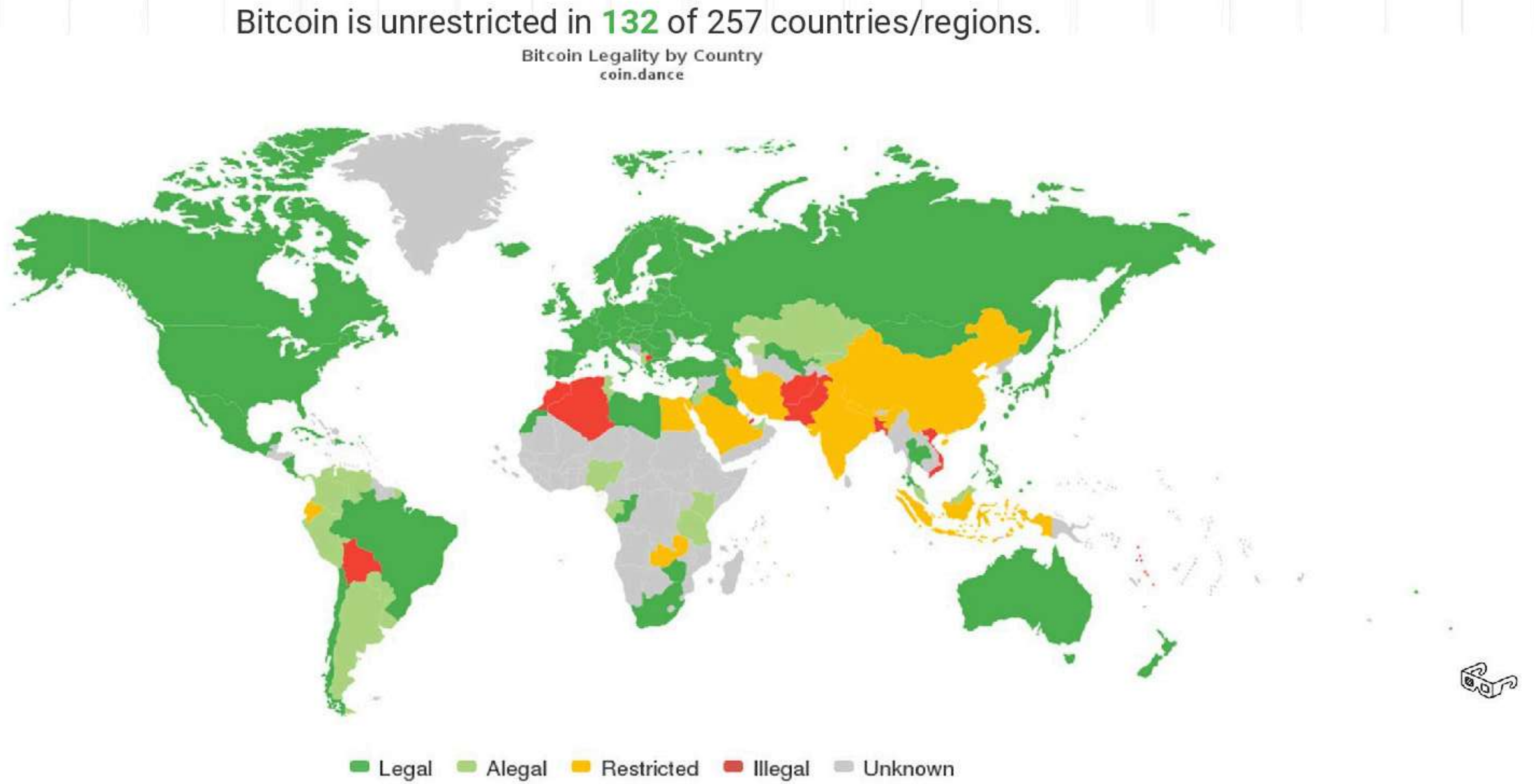
- Single Bitcoin Transaction Footprints
  - Carbon Footprint ***634.72kg*** CO_2
  - Electrical Energy ***1137.98*** kWh
  - equivalent of ***1406 760*** VISA transactions
  - Fresh Water Consumption ***17 935* L**



EFTA

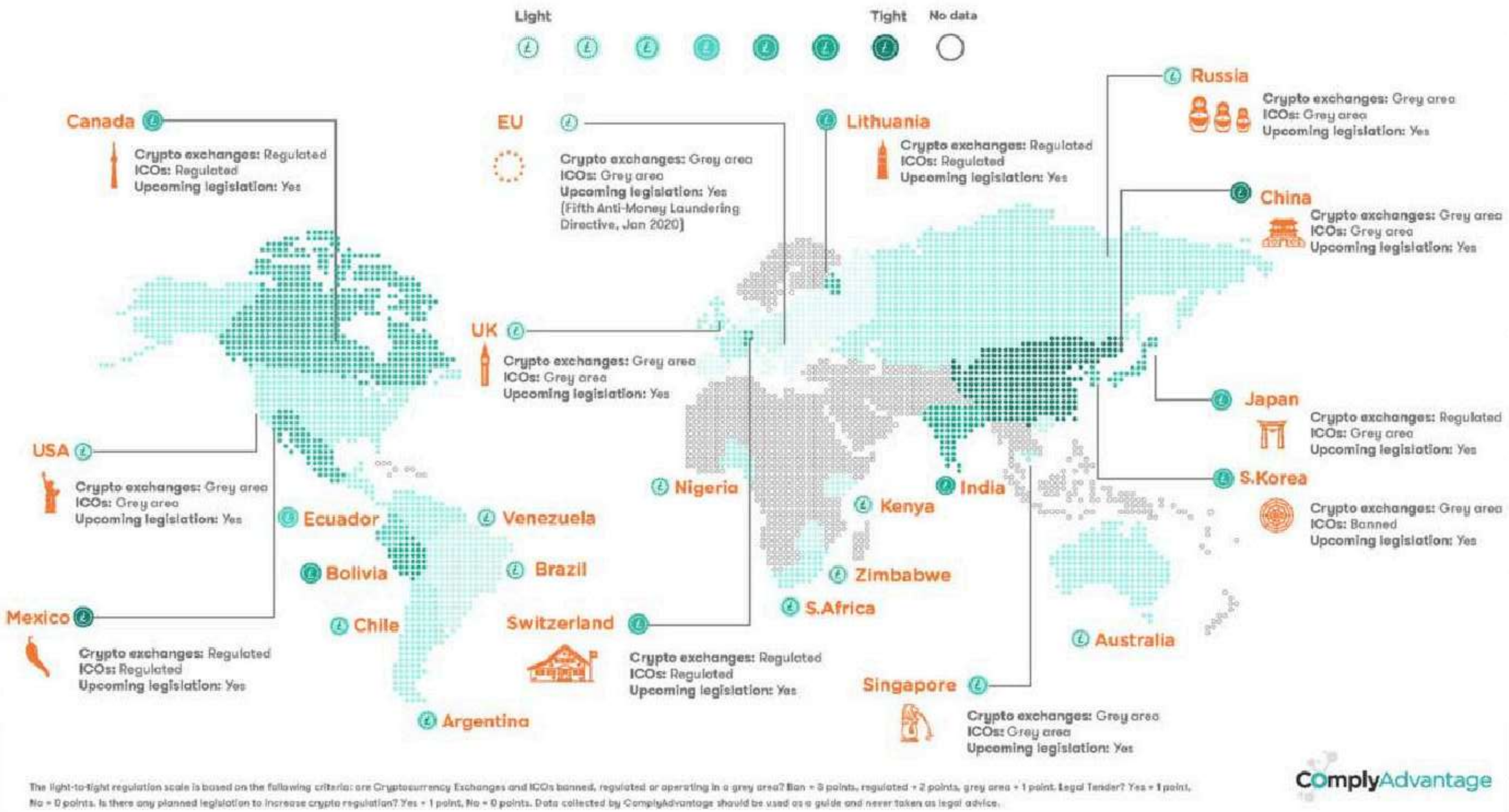
- ***Electronic Fund Transfer Act***
  - passed in US Congress in 1978
  - rights and liabilities of consumers
  - responsivities of everyone in activates related to electronic fund transfer
  - helps limits losses for lost / stolen funds (e.g. credit cards)

BTC legality in the world



Crypto Regulations by Country

How do different countries approach crypto regulations



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